Sharpness of the Dimension Bound for Benamou–Brenier Sparse Representations

Automated Math Discovery Run fa_banach_001, lane 4

9 August 2026

Abstract

Bredies, Carioni, Fanzon, and Romero proved that a dynamic inverse problem with finite-dimensional data H and Benamou–Brenier regularization has a minimizer representable by at most $\dim H$ trajectory atoms, and asked whether this upper bound is optimal. We show that it is optimal without additional assumptions. For every $D \geq 1$ and every $\alpha, \beta > 0$, we construct a one-sample problem with $H = \mathbb{R}^D$, a weak-star continuous observation operator, and squared Hilbert fidelity whose unique minimizer has atomic complexity exactly D.

1 Source question and answer

For a narrowly continuous positive measure curve $\rho = dt \otimes \rho_t$ and a momentum measure m satisfying the homogeneous continuity equation, the source uses

$$J_{\alpha,\beta}(\rho, m) = \beta B(\rho, m) + \alpha \|\rho\|_{\mathcal{M}((0,1)\times\overline{\Omega})}, \qquad \alpha, \beta > 0,$$

where B is the Benamou–Brenier kinetic energy. For a continuous linear observation A with values in a finite-dimensional Hilbert space H and a convex fidelity F, its representer theorem gives a minimizer that is a positive combination of at most $\dim H$ normalized trajectory atoms. The paper explicitly asks whether this number can be improved; see Figure 1.

Theorem 1 (Universal sharpness)**.** *For every integer $D \geq 1$ and every $\alpha, \beta > 0$, there is an instance of the source's discrete-time inverse problem with one sampling time, $H = \mathbb{R}^D$, a weak-star continuous observation operator, and fidelity $F(s) = \frac{1}{2}\|s - y\|_2^2$, such that:*

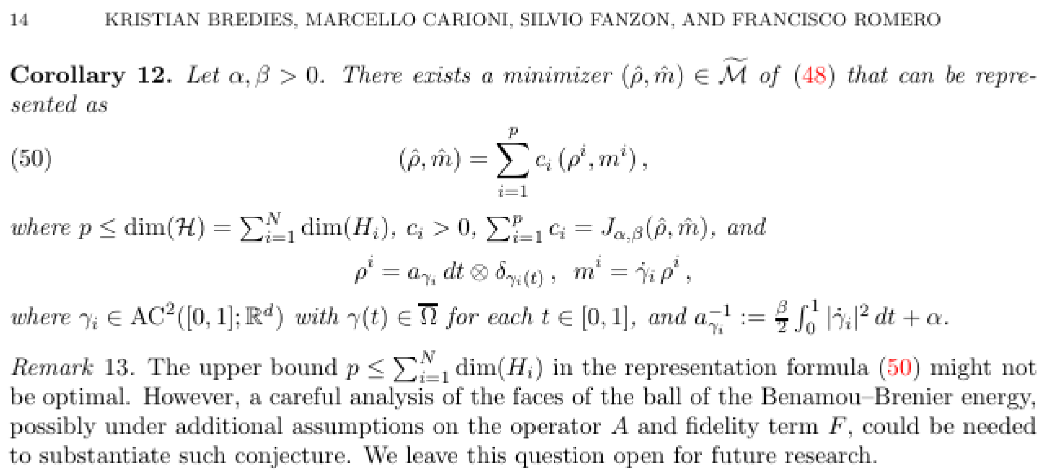
1. *the problem has a unique minimizer $(\widehat{\rho}, \widehat{m})$;*
2. *it has a representation by exactly D normalized trajectory atoms;*
3. *every representation of $(\widehat{\rho}, \widehat{m})$ as a positive combination of normalized trajectory atoms uses at least D atoms.*

Consequently, the general bound $p \leq \dim H$ in the source theorem and the bound $p \leq \sum_i \dim H_i$ in its discrete-time corollary cannot be decreased without extra hypotheses.

2 The measurement construction

Take the connected one-dimensional domain $\Omega = (0, 1)$ and fix a sampling time $t_0 \in (0, 1)$. Choose distinct points $x_1, \ldots, x_D \in \Omega$ and continuous functions $\varphi_j : \overline{\Omega} \to [0, 1]$ with pairwise disjoint supports such that

$$\varphi_j(x_j) = 1, \qquad \{x : \varphi_j(x) = 1\} = \{x_j\}.$$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Corollary 12. *Let $\alpha, \beta > 0$. There exists a minimizer $(\hat\rho, \hat m) \in \widetilde{\mathcal{M}}$ of (48) that can be represented as*

$$(\hat\rho, \hat m) = \sum_{i=1}^{p} c_i \,(\rho^i, m^i)\,, \tag{50}$$

where $p \le \dim(\mathcal{H}) = \sum_{i=1}^N \dim(H_i)$, $c_i > 0$, $\sum_{i=1}^p c_i = J_{\alpha,\beta}(\hat\rho, \hat m)$, and

$$\rho^i = a_{\gamma_i}\, dt \otimes \delta_{\gamma_i(t)}\,, \quad m^i = \dot\gamma_i\, \rho^i\,,$$

where $\gamma_i \in \mathrm{AC}^2([0,1];\mathbb{R}^d)$ with $\gamma(t) \in \overline{\Omega}$ for each $t \in [0,1]$, and $a_{\gamma_i}^{-1} := \frac{\beta}{2}\int_0^1 |\dot\gamma_i|^2\, dt + \alpha$.

Remark 13. The upper bound $p \le \sum_{i=1}^N \dim(H_i)$ in the representation formula (50) might not be optimal. However, a careful analysis of the faces of the ball of the Benamou–Brenier energy, possibly under additional assumptions on the operator A and fidelity term F, could be needed to substantiate such conjecture. We leave this question open for future research.

Figure 1: Corollary 12 and the open sharpness question in arXiv:1907.11589.

For example, sufficiently narrow triangular bumps around the x_j work. In particular,

$$\sum_{j=1}^{D} \varphi_j \le 1, \qquad \left\{ x : \sum_{j=1}^{D} \varphi_j(x) = 1 \right\} = \{x_1, \dots, x_D\}. \tag{1}$$

Let $H = \mathbb{R}^D$ and define the one-time observation

$$A\rho = \left(\int \varphi_1 \, d\rho_{t_0}, \dots, \int \varphi_D \, d\rho_{t_0} \right). \tag{2}$$

This satisfies exactly the source's continuity requirement: weak-star convergence of ρ_{t_0} implies convergence of all D integrals. Set

$$y = (\alpha + 1, \dots, \alpha + 1) \quad \text{and} \quad G(\rho, m) = J_{\alpha,\beta}(\rho, m) + \frac{1}{2}\|A\rho - y\|_2^2.$$

3 The unique minimizer

We first reduce the infinite-dimensional problem to a sharp scalar inequality.

Lemma 1. *If $J_{\alpha,\beta}(\rho, m) < \infty$ and $s = A\rho$, then $s_j \ge 0$ and*

$$J_{\alpha,\beta}(\rho, m) \ge \alpha \sum_{j=1}^{D} s_j. \tag{3}$$

Equality in (3) implies $m = 0$ and that the time slices are equal to one fixed positive measure supported on $\{x_1, \dots, x_D\}$.

Proof. Finiteness of the Benamou–Brenier integrand forces $\rho \ge 0$. Testing the continuity equation against spatially constant functions shows that $M := \rho_t(\overline{\Omega})$ is independent of t. Hence $\|\rho\|_{\mathcal{M}} = M$. By (1),

$$\sum_{j=1}^{D} s_j = \int \sum_{j=1}^{D} \varphi_j \, d\rho_{t_0} \le M,$$

and therefore

$$J_{\alpha,\beta}(\rho, m) = \beta B(\rho, m) + \alpha M \geq \alpha M \geq \alpha \sum_j s_j.$$

If equality holds, then $B(\rho, m) = 0$, so the velocity density and hence m vanish. The continuity equation becomes $\partial_t \rho = 0$, and narrow continuity gives $\rho_t = \mu$ for a fixed positive measure μ. Equality in the mass estimate says $\int (1 - \sum_j \varphi_j)\, d\mu = 0$. Formula (1) then forces $\operatorname{supp} \mu \subset \{x_1, \ldots, x_D\}$. $\square$

For any admissible pair and $s = A\rho$, Lemma 1 yields

$$G(\rho, m) \geq \sum_{j=1}^{D} \left[\alpha s_j + \frac{1}{2}(s_j - (\alpha + 1))^2 \right]. \tag{4}$$

On $[0, \infty)$ each bracket is strictly convex and has its unique minimum at $s_j = 1$. This lower bound is attained by

$$\widehat{\rho} = dt \otimes \sum_{j=1}^{D} \delta_{x_j}, \qquad \widehat{m} = 0, \tag{5}$$

because $A\widehat{\rho} = (1, \ldots, 1)$ and $J_{\alpha,\beta}(\widehat{\rho}, 0) = \alpha D$.

Now let (ρ, m) be any minimizer. Equality must hold both in (4) and in the scalar minimizations, so $s_j = 1$ for every j and Lemma 1 applies with equality. Thus $m = 0$ and $\rho_t = \mu$ with μ supported on the x_j. Since $1 = s_j = \int \varphi_j \, d\mu = \mu(\{x_j\})$, we obtain $\mu = \sum_j \delta_{x_j}$. This proves uniqueness of (5).

4 Exact atomic complexity

The normalized trajectory atom associated with an absolutely continuous curve γ is

$$\rho^\gamma = a_\gamma \, dt \otimes \delta_{\gamma(t)}, \qquad m^\gamma = \dot{\gamma} \, \rho^\gamma, \qquad a_\gamma^{-1} = \alpha + \frac{\beta}{2} \int_0^1 |\dot{\gamma}(t)|^2 \, dt.$$

Its observation vector is

$$A\rho^\gamma = a_\gamma \big(\varphi_1(\gamma(t_0)), \ldots, \varphi_D(\gamma(t_0)) \big). \tag{6}$$

Because the supports of the φ_j are disjoint, the vector in (6) has at most one nonzero coordinate.

Suppose

$$(\widehat{\rho}, \widehat{m}) = \sum_{\ell=1}^{p} c_\ell (\rho^{\gamma_\ell}, m^{\gamma_\ell}), \qquad c_\ell > 0.$$

Applying A writes $(1, \ldots, 1)$ as a sum of p vectors, each supported in at most one coordinate. Since all D coordinates of the sum are nonzero, necessarily $p \geq D$.

Conversely, let $\gamma_j(t) \equiv x_j$. Then $a_{\gamma_j} = 1/\alpha$, and

$$(\widehat{\rho}, \widehat{m}) = \sum_{j=1}^{D} \alpha (\rho^{\gamma_j}, m^{\gamma_j}).$$

Thus $p = D$ is attained. This completes the proof of Theorem 1.

5 Scope

The result establishes optimality of the source's bound over the full class of admissible observation operators and squared fidelities, already for one sampling time and a connected interval domain. It does not preclude smaller bounds under additional geometric assumptions coupling the measurement channels or under hypotheses that force a single trajectory to contribute to several independent data coordinates.

References

[1] K. Bredies, M. Carioni, S. Fanzon, and F. Romero, *On the extremal points of the ball of the Benamou–Brenier energy*, Bull. Lond. Math. Soc. 53 (2021), 1436–1452; arXiv:1907.11589.